1 I

1.1 (1)

From Gauss's law,

$$\nabla \cdot \vec{D} = \rho_{\text{free}}, \qquad \oint \vec{D} \cdot d\vec{S} = q.$$

By spherical symmetry,

$$D(r)\, 4\pi r^2 = q,$$

and therefore

$$\boxed{\vec{D}(r) = \frac{q}{4\pi r^2}\hat{r}}.$$

This expression holds both inside and outside the sphere (for $r \neq 0$).

1.2 (2)

Since $\vec{D} = \epsilon \vec{E}$ inside the dielectric, the electric field is

$$\vec{E}(r) = \begin{cases} \dfrac{q}{4\pi\epsilon r^2}\hat{r}, & 0 < r < R, \\ \dfrac{q}{4\pi\epsilon_0 r^2}\hat{r}, & r > R. \end{cases}$$

In both regions, the field points radially outward because $q > 0$.

1.3 (3)

The polarization is given by

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E}.$$

Thus,

$$\vec{P}(r) = \begin{cases} \dfrac{q}{4\pi r^2}\left(1 - \dfrac{\epsilon_0}{\epsilon}\right)\hat{r}, & 0 < r < R, \\ 0, & r > R. \end{cases}$$

The induced (bound) surface charge density is

$$\sigma_{\text{b}} = \vec{P}(R^-) \cdot \hat{n} = \boxed{\frac{q}{4\pi R^2}\left(1 - \frac{\epsilon_0}{\epsilon}\right)}.$$

Here $\hat{n} = \hat{r}$ is the outward normal to the dielectric surface. Although $\vec{P}(R^+) = 0$, the surface charge is determined by the polarization on the dielectric side.

The total induced charge on the surface is

$$Q_{\text{surface}} = 4\pi R^2 \sigma_{\text{b}} = q\left(1 - \frac{\epsilon_0}{\epsilon}\right).$$

Since the total bound charge must vanish, the induced charge at the center is

$$\boxed{Q_{\text{center}} = -Q_{\text{surface}} = -q\left(1 - \frac{\epsilon_0}{\epsilon}\right)}.$$

2 II

2.1 (4)

The dipole moment per molecule is

$$\vec{p}_m = \frac{\vec{P}}{N} = \epsilon_0 \alpha \vec{E}',$$

so that

$$\vec{E}' = \frac{\vec{P}}{\epsilon_0 N \alpha}.$$

From the Clausius–Mossotti relation,

$$\chi = \frac{N\alpha}{1 - N\alpha/3}, \qquad N\alpha = \frac{\chi}{1 + \chi/3}.$$

Substituting this into the expression for the local field gives

$$\begin{aligned} \vec{E}' &= \frac{\vec{P}}{\epsilon_0}\left(\frac{1}{\chi} + \frac{1}{3}\right) \\ &= \boxed{\vec{E} + \frac{\vec{P}}{3\epsilon_0}}, \end{aligned}$$

where $\vec{P} = \epsilon_0 \chi \vec{E}$ has been used.

2.2 (5)

The difference between the local field $\vec{E}'$ and the macroscopic field $\vec{E}$ arises from the polarization of the surrounding material. The macroscopic field is spatially averaged, whereas a molecule experiences an additional field due to nearby polarization charges. In the Lorentz spherical-cavity approximation, this contribution is $\vec{P}/(3\epsilon_0)$.

3 III

3.1 (6)

In the static equilibrium state, $\dot{\vec{x}} = \ddot{\vec{x}} = 0$. The equation of motion therefore becomes

$$-m\omega_0^2 \vec{x} - e\vec{E}' = 0, \qquad \vec{x} = -\frac{e}{m\omega_0^2}\vec{E}'.$$

The induced dipole moment is

$$\vec{p}_m = -e\vec{x} = \frac{e^2}{m\omega_0^2}\vec{E}' = \epsilon_0 \alpha \vec{E}'.$$

Hence,

$$\boxed{\alpha = \frac{e^2}{\epsilon_0 m \omega_0^2}}.$$

3.2 (7)

For an oscillating local field, assume

$$\vec{E}'(t) = \vec{E}'(\omega)e^{-i\omega t}, \qquad \vec{x}(t) = \vec{x}(\omega)e^{-i\omega t}.$$

Substituting into the equation of motion yields

$$-m\omega^2\vec{x} = -m\omega_0^2\vec{x} + i\lambda m\omega\vec{x} - e\vec{E}',$$

where the frequency-dependent amplitudes are understood. Thus,

$$\vec{p}_m(\omega) = -e\vec{x}(\omega) = \frac{e^2}{m(\omega_0^2 - \omega^2 - i\lambda\omega)}\vec{E}'(\omega).$$

Comparing this with $\vec{p}_m(\omega) = \epsilon_0\alpha(\omega)\vec{E}'(\omega)$, we obtain

$$\boxed{\alpha(\omega) = \frac{e^2}{\epsilon_0 m(\omega_0^2 - \omega^2 - i\lambda\omega)}}.$$

3.3 (8)

Multiplying the numerator and denominator by the complex conjugate,

$$\alpha(\omega) = \frac{e^2}{\epsilon_0 m}\frac{(\omega_0^2 - \omega^2) + i\lambda\omega}{(\omega_0^2 - \omega^2)^2 + (\lambda\omega)^2}.$$

Therefore,

$$\boxed{\operatorname{Im}\alpha(\omega) = \frac{e^2}{\epsilon_0 m}\frac{\lambda\omega}{(\omega_0^2 - \omega^2)^2 + (\lambda\omega)^2} > 0}.$$

3.4 (9)

From the Clausius–Mossotti relation,

$$\epsilon(\omega) = \epsilon_0[1 + \chi(\omega)], \qquad \chi(\omega) = \frac{N\alpha(\omega)}{1 - N\alpha(\omega)/3}.$$

In particular,

$$\operatorname{Im}\chi(\omega) = \frac{N\operatorname{Im}\alpha(\omega)}{|1 - N\alpha(\omega)/3|^2} > 0,$$

so $\operatorname{Im}\epsilon(\omega) > 0$.

For a plane wave propagating in the z-direction, let

$$\vec{E}(z,\omega) = \vec{E}_0 e^{ikz}, \qquad k^2 = \frac{\omega^2\epsilon(\omega)}{\epsilon_0 c^2}.$$

Since the permittivity is complex, the wave number is also complex:

$$k = k' + ik'', \qquad k'' > 0.$$

Using the time dependence $e^{-i\omega t}$, the field becomes

$$\vec{E}(z,t) = \operatorname{Re}\left[\vec{E}_0 e^{-k''z}e^{i(k'z-\omega t)}\right].$$

Hence, its amplitude decreases as $e^{-k''z}$ during propagation. A positive imaginary part of the polarizability therefore corresponds to attenuation of the electromagnetic wave.